Milestone 3 (Results and Evaluation) — Research Project

Performance Analysis of Multi-Segment Pinching-Antenna ISAC Under LoS Blockage

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Abstract

Scope. The project analyzes a multi-segment pinching-antenna system (PASS) that performs integrated sensing and communication (ISAC) under environment-division multiple access (EDMA): each waveguide segment activates one antenna at the midpoint of its user and target, line-of-sight (LoS) blockage isolates concurrently served segments, and closed-form lower bounds on the ergodic sum communication rate (CR) and sensing rate (SR) were derived and verified at Milestone 2 (M2). This milestone (M3) runs the full numerical evaluation against transmit power P and the number of segments M, for three waveguide lengths, under a physically grounded automated-guided-vehicle (AGV) target radar cross-section (RCS) and a conventional uniform linear array (ULA) benchmark, both of which M2 had deferred. EDMA nearly triples the sum CR of time-division multiple access (TDMA) and raises the sum SR up to sevenfold; the ULA trails both. The high-SNR design rule for choosing M is validated, including where it is only a necessary and not a sufficient condition, and the state-of-the-art (SOTA) positioning is revised accordingly.

1 Model Recap

A base station feeds a lossless waveguide of length D_L, split into M isolated segments of width $S = D_L/M$, each with its own feed. Segment m serves a user U_m and senses a target T_m, both uniform within the segment, through one pinching antenna PA_m at their midpoint. A link of length r is unblocked with probability $e^{-\phi r^2}$; interference, clutter, and leakage reach segment m only through unblocked cross links from the two adjacent segments. The communication SINR and achievable rate, and the sensing SINR and rate, are

$$\gamma_m = \frac{P|h_{m,m}|^2}{\sigma^2 + P\sum_i \delta_{i,m}|h_{i,m}|^2}, \quad \Gamma_m = \frac{\rho_s \mu_{m,m}}{1 + \rho_s \sum_i \delta^{\mathrm{s}}_{m,i}(\mu_{m,i} + \nu_{m,i}/L)}, \tag{1}$$

with $R_m = \log_2(1+\gamma_m)$ and $R_{s,m} = \frac{1}{L}\log_2(1+\Gamma_m)$; $\mu_{m,i}$, $\nu_{m,i}$ are the clutter and leakage gains, and the target reflection follows $\beta \sim \mathcal{CN}(0, \alpha_s)$. M2 derived the closed-form cross-link LoS probability $p_1(\phi)$ and, via Boole's inequality, the lower bounds

$$\bar{R}_c \geq \sum_{m=1}^{M} \left[B_{\mathrm{c}} - B_{\mathrm{c}}^{\max}|\mathcal{N}_m^{\mathrm{nb}}|p_1(\phi)\right]^+, \quad \bar{R}_s \geq \sum_{m=1}^{M} \left[B_{\mathrm{s}} - B_{\mathrm{s}}^{\max}|\mathcal{N}_m^{\mathrm{nb}}|p_1(\phi)\right]^+. \tag{2}$$

TDMA (one RF chain, power MP, no interference) is the multiple-access baseline; EDMA beats it at high SNR iff $p_1(\phi) < \frac{1}{2}$.

New in this milestone. (i) $\alpha_s = 4\pi\bar{\sigma}/\lambda^2$ is tied to a physical RCS $\bar{\sigma}$ instead of an arbitrary constant; the evaluation uses an AGV, $\bar{\sigma} = -11.235$ dBsm, giving $\alpha_s = 8237$. (ii) A conventional, non-reconfigurable M-element half-wavelength ULA at the feed end, with all links LoS, is added as a baseline through the same SINR forms (1).

2 Evaluation Setup

No external dataset is used; every point is from a Monte Carlo (MC) simulator verified against every closed form, averaging 10^5 random user/target placements with the blockage expectation evaluated exactly (all cross-link patterns enumerated). Fixed parameters: $f_c = 28$ GHz, $d = 3$ m, $\sigma^2 = \sigma_s^2 = -90$ dBm, $L = 5$, $\phi = 0.02$ m^{-2}, $\bar{\sigma} = -11.235$ dBsm, $P \in [0, 30]$ dBm, $D_L \in \{10, 20, 40\}$ m. Three schemes are compared under matched placements/blockage draws: EDMA (midpoint), TDMA, and the ULA; an ablation lets every segment interfere (not only the two adjacent) to test the approximation behind (2).

3 Results

Fig. 1 shows both bounds below the MC curve at every P: tight for $M = 2$ (CR 29.24 vs. 29.45; SR 2.22 vs. 2.28 bits/s/Hz at $P = 30$ dBm) and looser for $M = 5$ (CR 43.61 vs. 51.43, SR 3.83 vs. 5.61). Including every segment, not just the two adjacent ones, changes the CR and SR by at most 1.1% and 0.4%, supporting the adjacent-only approximation. EDMA leads TDMA by 1.85–2.85× in CR ($M = 2, 5$) and 2.7–7.1× in SR;

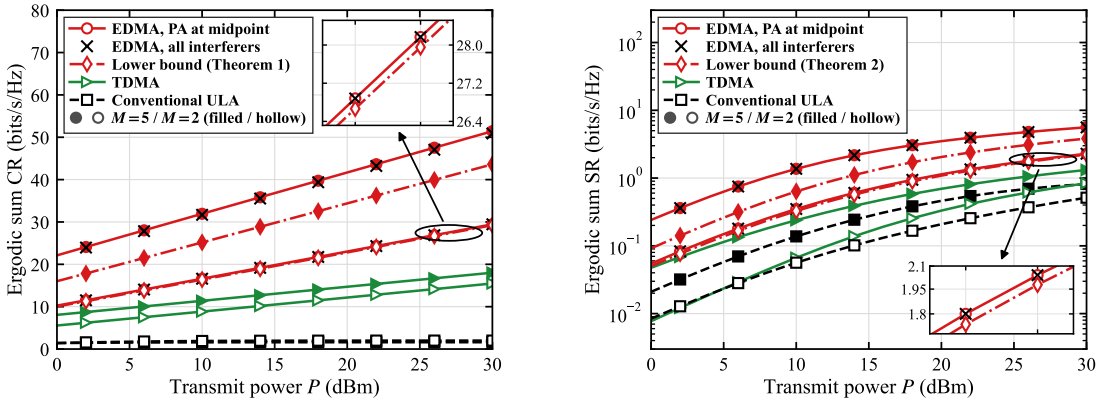


Figure 1: Sum CR and SR vs. P, $D_L = 40$ m (hollow: $M = 2$, filled: $M = 5$). Crosses: all segments interfere. Diamonds: lower bound.

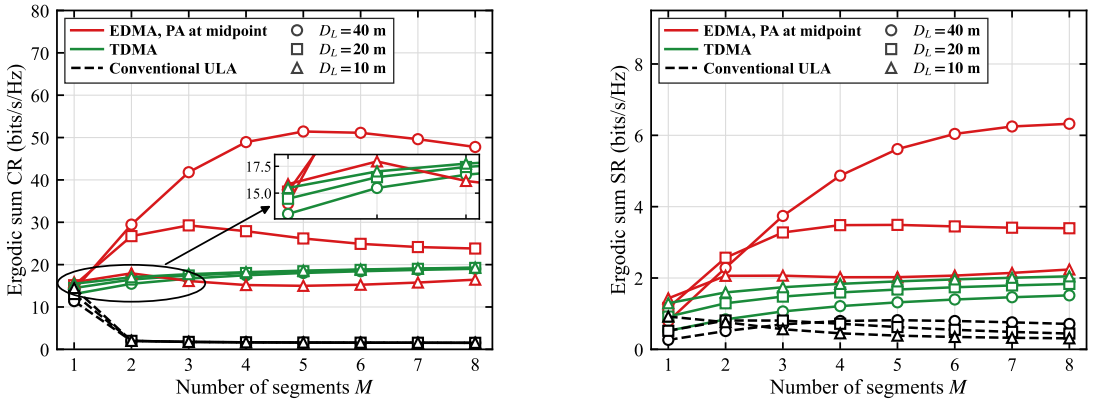


Figure 2: Sum CR and SR vs. M at $P = 30$ dBm, $D_L \in \{10, 20, 40\}$ m.

the ULA trails both throughout, since its co-located elements interfere with each other and sit far from most users/targets.

Fig. 2 sweeps M from 1 to 8. The CR peaks then falls as narrower segments raise $p_1(\phi)$: 51.4 at $M = 5$ ($D_L = 40$ m), 29.3 at $M = 3$ ($D_L = 20$ m), 18.0 at $M = 2$ ($D_L = 10$ m); the best M grows with D_L. The SR never falls: it rises to 6.32 bits/s/Hz by $M = 8$ for $D_L = 40$ m and saturates (dip $\leq 2.7\%$) for the shorter waveguides, since the own-link round trip keeps dominating clutter and leakage. The ULA stays below EDMA and TDMA for every M, D_L (at most 94% of TDMA's CR, 71% of its SR).

4 Critical Analysis

- **Design rule.** $p_1(\phi) < \frac{1}{2}$ correctly flags the only CR failure case: for $D_L = 10$ m, $M \geq 3$, $p_1(\phi) \geq 0.61$ and EDMA's CR falls to 0.81–0.91× TDMA's. For $D_L = 20$ m, $p_1(\phi)$ exceeds $\frac{1}{2}$ already at $M = 4$, yet EDMA still leads TDMA for every $M \leq 8$ (min. ratio 1.046): the rule bounds only the high-SNR *slope*, so it is necessary but not sufficient away from its own boundary.
- **Bound tightness.** The CR gap grows from $\sim 0\%$ ($M = 2$) to $\sim 15\%$ ($M = 5$); the SR gap from $\sim 3\%$ to $\sim 32\%$, since the union bound $\Pr\{\mathcal{A}_m\} \leq |\mathcal{N}_m^{\text{nb}}| p_1$ loosens as p_1 grows. The bound is exact at $M = 1$ (within 0.1%, no cross links) and, at $\phi \to 0$, correctly collapses to 0 while EDMA itself falls below TDMA (e.g., 8.57 vs. 15.48, $M = 2$).
- **Ablation.** Restricting interference/clutter/leakage to the two adjacent segments, instead of all $M - 1$, costs at most 1.1% CR / 0.4% SR, confirming non-adjacent links are weak and usually blocked.

5 Revised SOTA Positioning

M2 positioned the project on capability alone. M3 adds a quantitative answer that none of the surveyed works give: how much a reconfigurable antenna is worth against a conventional array under the same blockage model (the ULA never exceeds 94%/71% of TDMA's CR/SR, let alone EDMA's), now stated against a concrete deployment target (the AGV RCS) rather than an arbitrary reflection strength.

Table 1: Capability comparison with the closest prior work.

Work	PASS	ISAC	Blockage	Concurrent segments	Non-PASS baseline
Ding & Poor [1]	✓	×	✓	×	×
EDMA [2]	✓	×	✓	✓	×
Ouyang *et al.* [3]	✓	✓	×	×	×
SWAN [4]	✓	✓	×	✓	×
This project	✓	✓	✓	✓	✓

6 Conclusion

The sum CR and SR were evaluated against P and M for three D_L under a physically grounded AGV RCS; EDMA was compared against TDMA and a conventional ULA; the adjacent-only approximation was checked by ablation; and the $p_1(\phi) < \frac{1}{2}$ design rule was validated, including where it is only necessary. EDMA nearly triples TDMA's sum CR and raises its sum SR up to sevenfold, the ULA trails both, and the bounds remain valid everywhere tested while loosening exactly where EDMA itself nears its failure regime. The RCS and ULA items planned for M4 at the M2 stage are already evaluated here, so the remaining M4 work is the final write-up and submission package.

Individual contribution. All work in this single-member project was carried out by Khushi Shah under the mentorship of Prof. Dhaval Patel.

References

[1] Z. Ding and H. V. Poor, "LoS blockage in pinching-antenna systems: Curse or blessing?" *IEEE Wireless Commun. Lett.*, vol. 14, no. 9, pp. 2798–2802, 2025.

[2] Z. Ding, R. Schober, and H. V. Poor, "Environment division multiple access (EDMA): A feasibility study via pinching antennas," *IEEE Trans. Wireless Commun.*, vol. 25, pp. 15 675–15 691, 2026.

[3] C. Ouyang, Z. Wang, Y. Liu, and Z. Ding, "Rate region of isac for pinching-antenna systems," *IEEE Trans. Commun.*, vol. 74, pp. 5849–5866, 2026.

[4] H. Jiang, C. Ouyang, Z. Wang, Y. Liu, A. Nallanathan, Z. Ding, and R. Schober, "Segmented waveguide-enabled pinching-antenna systems (SWANs) for ISAC," arXiv:2512.07649, 2025.